

1 **Metagenomic Analysis of Fecal Samples as** 2 **a Differential Diagnostic Tool for** 3 **Colorectal Cancer**

4 **Kaitlyn Schisler,^{1*} Cory Sporn,¹ Faith Shipman,¹ Jacob Anderson¹**

5 ¹New College of Interdisciplinary Arts and Sciences, Arizona State University,
6 Tempe, Arizona, USA

7 * Address correspondence to Kaitlyn Schisler, kschisl1@asu.edu.

8 Author order was selected through drawn straws.

9 **ABSTRACT**

10 Colonoscopies are an accurate screening method for colorectal cancer (CRC),
11 which is on the rise globally. Some individuals prolong or decline screenings due
12 to their invasive nature, requiring other accurate diagnostic tools. Metagenomics
13 has become more accessible, with large datasets being curated from multiple
14 countries. This investigation used the curatedMetagenomicData version3 (cMD3)
15 to identify studies associated with CRC and polyps to build a random forest (RF)
16 model to classify individuals as a positive or negative case for adenomatous or
17 cancerous polyps. This meta-analysis totaled 1,643 samples from eleven studies,

18 which included patient metadata, relative abundance, and KEGG pathway
19 abundance and coverage. Using a study-aware data split (80% training, 20%
20 testing), the model's accuracy was 64.89% ($p = 0.0026$, $F1 = 0.7268$, $AUC = 0.7170$).
21 Leave-one-dataset-out (LODO) cross-validation models demonstrated moderate
22 cross-study performance with high sensitivity but reduced specificity, indicating
23 limited generalizability across heterogeneous cohorts. Of eleven LODO models,
24 five showed significant p-values at a 95% confidence level ($F1 = 0.6806$ – 0.8081 ,
25 $AUC = 0.6970$ – 0.9550), demonstrating the variability in model performance when
26 training data does not adequately represent the test distribution, particularly with
27 underrepresented populations. Fecal Occult Blood Tests (FOBTs) metrics were
28 compared to determine diagnostic readiness. Ultimately, a high volume of false
29 positives is expected in comparison to FOBTs, which is far from ideal. While some
30 metrics outperform FOBTs, the RF models do not have the specificity required,
31 suggesting the need for more data and refinement before development and use as
32 a patient-ready diagnostic tool, though results are promising for clinical
33 metagenomics in the future.

34 **IMPORTANCE**

35 The human gut microbiome plays a role in immunity, metabolism, and overall
36 human health. Only in recent years have scientists begun to fully understand the
37 complexity of the gut microbial ecosystem. Every individual has a unique
38 microbiome fingerprint that changes over time with aging, dietary shifts, and
39 environmental or geographic influences. Studies have also shown that the
40 microbiome performs biochemical functions that can impact disease diagnosis and
41 treatment strategies [1]. Accurate clinical decision making depends on the
42 integration of all available patient information, meaning that microbiome-based

43 insights would provide an additional layer of evidence. A tool like this would
44 support clinicians' differential diagnosis and treatment in a timely, non-invasive
45 manner.

46 INTRODUCTION

47 Cancer is the second highest cause of death worldwide and in the United
48 States (U.S.), with over 10 million deaths worldwide in 2020 and over 18 million
49 cases in 2022 [2–5]. Among these cases, colorectal cancer (CRC) has the third
50 highest incidence at approximately 1.9 million new cases and 904 thousand deaths
51 in 2022. The countries most impacted include China, Japan, and the US [5, 6].
52 There is a bottleneck with data collection and processing, leading to a 2 to 3 year
53 delay in reporting cancer incidence and mortality rates throughout the world.
54 Subsequently, Siegel et al. [3] modeled 2025 cancer impacts in the U.S. using prior
55 year rates as predictors, with their analyses resulting in over 2 million new cancer
56 cases predicted. More than 154 thousand belong to CRC, an estimated 53
57 thousand resulting in death, meaning that CRC would be the second leading
58 cause of cancer related deaths in the U.S. Furthermore, the incidence rate of CRC
59 is increasing as those born after 1950 approach higher risk ages [3], increasing the
60 burden on the healthcare system. Cancer, particularly CRC, continues to be a
61 fundamental target for prevention, diagnosis, and treatment.

62 Cancer screening remains one of the best procedures for positive patient
63 outcomes, especially because CRC is typically asymptomatic in early stages of the
64 disease; early detection often results in better prognosis [2]. This is reflected in the
65 2018 update of the CRC screening guideline, which reduces the recommended age
66 limit to 45 years from 55 years. As a result of this guideline change, there was a

67 three-fold increase in health insurance claims data between January 2021 and
68 December 2022 related to CRC screenings [3]. However, the COVID-19 pandemic
69 had the unfortunate side effect of reduced cancer screenings as hospitals had
70 limited resources, shifting priorities, and guidelines for reducing virus exposure,
71 as well as patients' loss of employment and health insurance. Overall, cancer
72 screenings, CRC diagnosis, and CRC-related surgeries remain reduced even after
73 the lifting of the quarantine. With the loss of >15 million preventative procedures
74 during the pandemic, there are predictions of 4000–7000 more deaths by CRC in
75 the U.S. than original projections, particularly affecting minority groups [3, 7].

76 A top priority to reduce the impact of these losses is to dissipate the backlog of
77 preventative care. Additional colonoscopy screenings per month would be helpful
78 [7], however, resource availability suggests a need for tools to determine priority
79 level in a rapid, non-invasive way. In addition to other methods such as fecal
80 occult blood tests (FOBT) and tumor markers [2, 8], a new and accurate screening
81 method would be highly beneficial in reducing the burden on colonoscopy suites
82 and improving differential diagnosis, even beyond the purposes of this backlog
83 catch-up. The human microbiome is a prime candidate for such a purpose, as
84 samples used for current non-invasive screenings like FOBT could also be
85 sequenced for microorganisms.

86 The human gut microbiome has a well known effect on host health, as
87 organisms inhabiting the gastrointestinal (GI) tract can be harmful, helpful, or
88 inert [9–11]. With microbes having a large influence on the nutrient metabolism,
89 synthesis of or interaction with molecules, and roles in disease protection, humans
90 certainly do not function optimally without them [12–14]. Prior epidemiological
91 and scientific evidence has also suggested that colon health and cancer risks are
92 affected by colonic microbial metabolism [15]. For example, the bacterial family

93 Lachnospiraceae is suspected to inhibit oral microbes from transitioning to the gut,
94 as oral-microbe enrichment in the GI tract is known to be disease associated,
95 including with CRC [16, 17]. On the other side, several microbial species have a
96 pathogenic effects, such as *Bacteroides fragilis*, *Fuseobacterium nucleatum*, and
97 *Escherichia coli*, which can release toxins and cause inflammation, potentially
98 leading to tumorigenesis [17–20]. The gut microbiome exists in dynamic
99 equilibrium, known as homeostasis, with its disruption known as dysbiosis
100 [17, 20]. Since the gut microbiome can shift between these states, there is interest
101 in identifying and characterizing the organisms present and their relative
102 abundances, called microbial fingerprinting. The microbiome can be characterized
103 in two main ways: the taxonomic profile (what species are present) and the
104 functional profile (what the species can do).

105 Through a retrospective, multi-cohort study of metagenomic human gut
106 microbiome data, our aim is to generate a supervised machine-learning (ML)
107 model with potential clinical applications for detecting the progression of CRC. By
108 investigating the predictive value of a microbial fingerprint for CRC progression,
109 we will generate the foundation of a proof-of-concept diagnostic tool to ease the
110 burden of colonoscopy suites. Based on the foundational work of others, our
111 hypothesis is that there will be a detectable difference in the microbial fingerprint
112 of different stages of CRC development.

113 It is well established that ML algorithms have been successful for diagnostics
114 and detecting disease biomarkers across a wide range of diseases, including CRC
115 [11, 16, 21]. Random Forest (RF) models in particular are optimal for disease
116 classification with minimal preprocessing, which is ideal for our use [21]. There is
117 also evidence that multi-kingdom analysis, including bacteria and archaea, results
118 in more accurate models according to AUC metrics [19]. However, since these

species are in relatively low abundance in the human gut, removal is likely to occur during prevalence filtering. Regardless, any remaining taxa will be included in a multi-kingdom model to increase model accuracy.

MATERIALS AND METHODS

Inclusion criteria for the study cohorts are based on CRC progression. Polyps and adenomas are benign growths with a high chance of becoming cancerous (15–40% and 2–40%, respectively), indicating relevance for CRC tracking [2]. Our multi-cohort study focused on benign and cancerous growths and is divided first into the classes described in Table 1.

Disease Class	Acronym	Description
Healthy Control	HC	Individuals with no diagnosed disease
Polyp/Adenoma	PA	Individuals with benign colorectal polyps or adenomas
Polyp/Adenoma + Comorbidities	PA+	Individuals with polyps/adenomas and additional conditions
Polyp/Adenoma + Metastases	PA-M	Individuals with polyps/adenomas with metastases
Colorectal Cancer	CRC	Individuals diagnosed with colorectal cancer
Colorectal Cancer + Comorbidities	CRC+	CRC patients with additional conditions
Colorectal Cancer + Metastases	CRC-M	CRC patients with metastases
History of CRC (Resection)	CRC-H	Individuals with prior CRC treated by surgical resection
Other Diseases	Other	Individuals with diseases unrelated to CRC progression

TABLE 1 Disease classification schema used in this study

Only stool samples are considered for model generation for data availability and future testing consideration. With age as a major factor in microbiome variation and CRC risk, we restricted samples to individuals over 18 years. Keeping individuals ≥ 18 years old rather than ≥ 45 years old, the recommended age to start screening, was justified because (i) there is evidence that incidence

134 rates are increasing in younger populations year over year [22], and (ii)
135 symptomatic individuals 20 years or older are suggested to be screened using
136 other methods [2]. Ideally, the diagnostic platform would be generalized to all
137 ages affected by CRC, especially as risks are increasing.

138 P. Li et al. [21] demonstrates that the performance of microbiome-based
139 disease diagnostic classifiers depends heavily on data processing and model
140 selection. Manghi et al. [16] generated a manually curated set of metagenomic
141 data with accompanying metadata, in which the information is available in a
142 standard form through Bioconductor (Supplementary Materials 8-B). Much of the
143 processing is completed, such as determining marker and pathway
144 abundance/coverage through HUMAnN3 and MetaPhlAn via bioBakery3,
145 known to reduce ambiguity and improve functional mapping [23]. This dataset,
146 curatedMetagenomicData version3(cMD3), was quintessential to jump start our
147 analysis, containing 22,710 samples from 94 datasets in 42 countries [16]. The
148 following parameters were applied to the full cMD3 dataset using R: (i) sample
149 site is stool, (ii) age is ≥ 18 years, (iii) the disease field was not blank, (iv) each
150 study's unique disease list contained text regarding CRC or polyp/adenoma, or
151 (v) the subject_id is unique (to avoid pseudoreplication).

152 This subset was expected to reflect many of the representation biases present in
153 cMD3 as well as overarching public data trends, dominated by samples from the
154 U.S., China, and European countries, with little representation from regions such
155 as Africa, South America, the Middle East, and much of Asia [11, 16]. With the
156 main contributor to an individual's microbiome being geographical location and
157 diet, this likely had a large influence on the data [10, 24, 25].

158 Initial exploration of the quantitative and qualitative data through descriptive
159 and inferential statistical analysis was conducive to detecting microbiome shifts
160 and guiding preprocessing prior to modeling. Analyses conducted were: (i) beta
161 diversity analysis for between-group variation using Bray–Curtis dissimilarity
162 [10, 23]; (ii) statistical tests to determine significant differences in alpha and beta
163 diversity as well as between categorical groups such as CRC progression classes,
164 including the Wilcoxon rank-sum test, permutational multivariate analysis of
165 variance (PERMANOVA) and ANOVA, and Linear Discriminant Analysis Effect
166 Size (LEfSe) [10, 21]; (iv) visualization techniques such as principal component
167 analysis (PCA), principal coordinates analysis (PCoA), box plots, bar plots, and
168 heat maps, revealing hidden patterns and effectively present findings from
169 statistical analyses [10, 21, 23].

170 It is well established that ML algorithms, and particularly RF, have been
171 successful for diagnostics and detecting disease biomarkers across a wide range of
172 diseases, including CRC [11, 16, 21]. This supervised learning technique allows
173 classification into healthy and diseased categories through a binary testing system,
174 effectively captures feature relationships, and is suitable for large datasets, making
175 it an appropriate model for CRC progression identification through microbial
176 fingerprinting [10, 19, 23].

177 Modeling began by combining the study data into a unified feature matrix that
178 included taxonomic abundance, pathway abundance, and pathway coverage data.
179 During the exploratory data analysis phase, a few rare disease subclasses were
180 identified and subsequently removed at this point with the goal of reducing noise,
181 namely CRC-M, PA-M, and CRC-H. Initial tests with the retention of HC, PA and
182 CRC separated yielded unsatisfactory results such as 0% identification of the PA

class, and ultimately a decision was made to re-bin with the RF classes found in Table 2.

	Random Forest Classes	Included CRC Classes
NEGATIVE_POLYP		HC, Other
POSITIVE_POLYP		PA, PA+, CRC, CRC+

TABLE 2 Final model classifiers for binary system.

After initial exclusions and binning was applied, modeling commenced. Two final models were built on the Sol high performance computing cluster at Arizona State University [26], initially with the randomforest package [27], with final iterations using the ranger package to improve modeling speed [28]. Model 1 was an 80/20 study-aware stratified train-test split, used for hyperparameter tuning due to processing speed and resource limitations. Model 2 was a leave-one-dataset-out (LODO) cross-validation, used for evaluation on completely unseen cohorts to better reflect real-world generalization and reduce the risk of overfitting to batch effects, providing a more robust assessment of model performance across heterogeneous datasets.

Data preprocessing before integration into the RF model was the same between Models 1 and 2 after the data split. Feature transformations were implemented on the training data, which were then blindly applied to the testing data to prevent data leakage in the. A prevalence threshold was applied to remove sparse features less than or equal to 1%, after which percentile normalization was performed to correct batch effects using HC as a reference distribution, enabling cross-study integration. While other optimal batch effect mitigation strategies were investigated, like MMUPHin, ComBat, and log-transformations, Gibbons et

204 al. [29] highlighted percentile normalization as a better solution for the correction
205 of batch effects in microbiome data, corroborated by our experimentation. The
206 generated percentile-normalized feature matrix was used as the input for the
207 model. To further improve class balance, class weighting was conducted,
208 calculated by an inverse frequency weighting. In the code, this is represented by
209 $\text{sum}(\text{class_counts})/(\text{length}(\text{class_counts})*\text{class_counts})$, which simplifies to $\frac{N}{K*n_i}$,
210 where N is the total number of samples, K is the number of classes, and n_i is the
211 number of samples in each class i . Special considerations were made for Model 2,
212 as the preprocessing was repeated within each LODO loop to ensure
213 normalization and balancing was per fold.

214 Each model's performance was evaluated using accuracy, a confusion matrix,
215 sensitivities, precision, F1 score, and area under the curve (AUC). Overfitting was
216 also assessed by a comparison between testing and training accuracy. Cross-study
217 performance was evaluated using both study level accuracy and Kappa scores.

218 The final tuning of models were implemented including variations in the
219 number of trees, minimum samples per terminal node, and prevalence filtering,
220 among others. Results of parameter changes were tracked via spreadsheet for
221 analysis, see Supplementary Materials 4. The final parameters that produced the
222 best results are listed in Table 3.

Parameter	Setting
Number of Trees	3000
Minimum Node Size	10
Max Tree Depth	10
Mtry Fraction	0.05
Mtry Size	493
Split Rule	"gini"
Prevalence Filter (%)	1%

TABLE 3 Final model parameters after tuning.

224 **RESULTS**

225 **Data Curation**

226 To understand the relationship between CRC, PA, or other combinations and
 227 the gut microbial community, we utilized data previously cataloged and profiled
 228 using cMD3 from the Waldron Lab. Dataset filtration resulted in a curated subset
 229 of 11 studies to be included in the model, totaling 1,643 samples; 617 HC and 1,026
 230 with an indicated disease (Table 4). The CRC-relevant studies included in this
 231 analysis contains between 59 and 616 samples, with the largest cohort being
 232 YachidaS_2019 and the smallest being ThomasAM_2018b. These studies span
 233 multiple geographic regions across Asia, Europe and North America, providing a
 234 diverse representation of populations relevant to CRC research. All studies
 235 include taxonomic relative abundance profiles, which contain 292 to 718 non-zero
 236 taxa per study. Functional metagenomic information is also available in the form
 237 of gene family, marker gene, and pathway abundance tables, though we focus

only on pathway abundance and coverage. These datasets vary substantially in size, reflecting the high dimensionality of functional genomic profiling from metagenomic sequencing.

Study	Samples	Country	Relative Abundance	Pathway Abundance	Pathway Coverage
YachidaS_2019	616	JPN	718	31291	30038
ZellerG_2014	156	FRA	652	22620	21492
FengQ_2015	154	AUT	606	22131	20879
YuJ_2015	128	CHN	575	21700	20636
WirbelJ_2018	125	DEU	537	17224	16341
VogtmannE_2016	104	USA	539	17940	16937
HanniganGD_2017	81	USA	292	8585	8103
ThomasAM_2018a	80	ITA	477	17388	16343
ThomasAM_2019_c	80	JPN	519	20391	19387
GuptaA_2019	60	IND	308	12535	12037
ThomasAM_2018b	59	ITA	501	16898	15981

TABLE 4 Summary table of the 11 available studies, sorted in descending order by sample size. Counts are non-zero features per study.

Along with the relative abundance and pathway data, there are 46 associated metadata points; several important demographic features like age and BMI are present, as well as clinical measurements like FOBT results, cholesterol, creatinine, and eGFR. However, when plotted for completeness (Figure 1), it becomes obvious that much of this data is incomplete for the vast majority of the observations in the sample set.

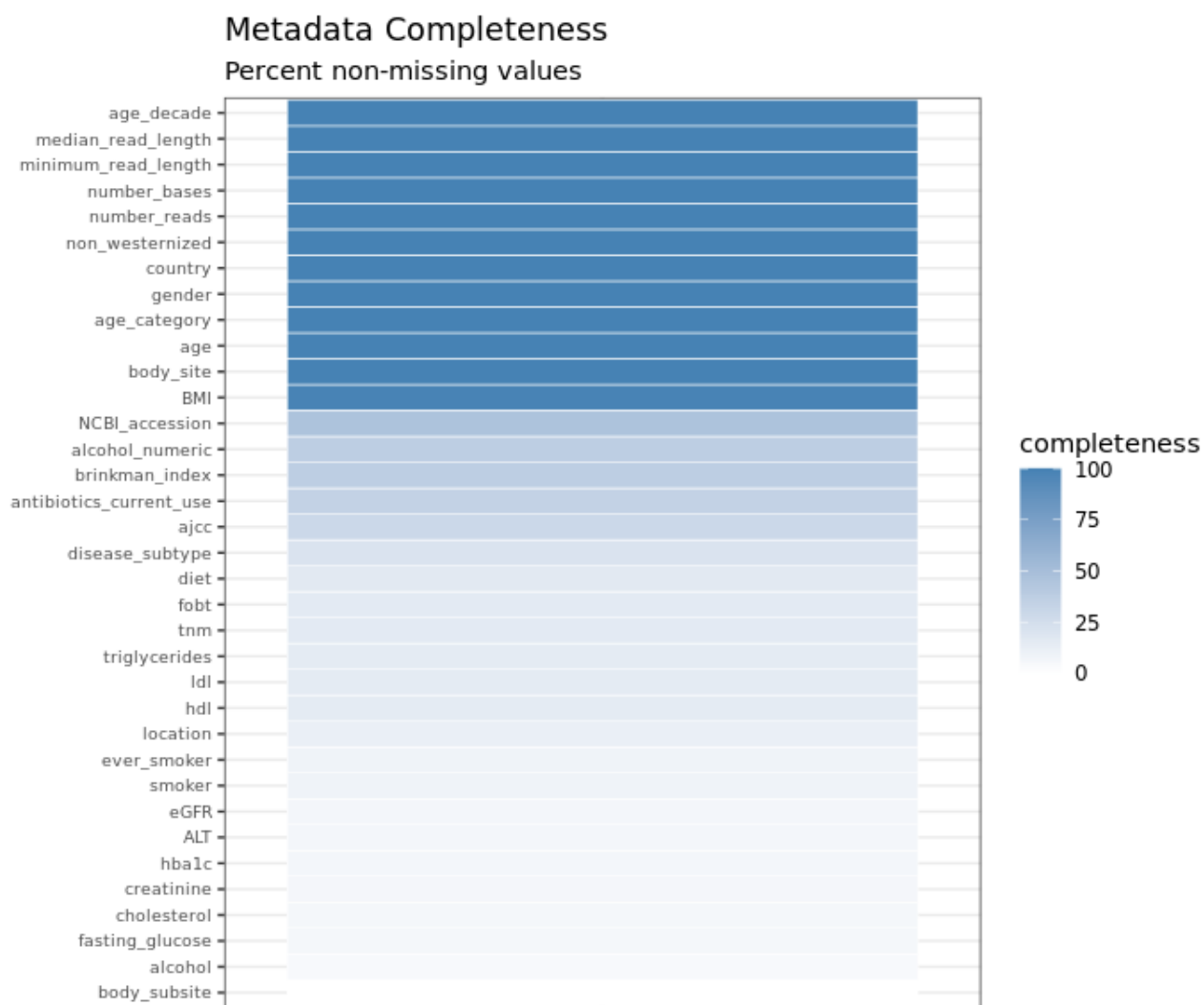


FIG 1 Metadata availability for selected studies.

249 The spread of the CRC progression classifiers was found to be quite
 250 imbalanced (Table 5, bar graph in Supplementary Materials 2-D), with HC and
 251 CRC being roughly equally balanced, and the other categories being starkly lower
 252 in counts, particularly CRC-M and PA-M. This contributed to the decision to drop
 253 and bin particular categories into the RF classifiers in the model.

	CRC Classifier	Count
	HC	617
	PA	164
	PA+	48
254	PA-M	2
	CRC	625
	CRC+	75
	CRC-M	1
	CRC-H	40
	Other	71

TABLE 5 Distribution of CRC progression classifiers. The bold rows were dropped from the final RF model. CRC-H was dropped as it is not a progression stage; it is unknown when the polyp was resected nor the time since resection, and thus the data points were removed to eliminate bias.

255 **Metadata Analysis**

256 The age summary statistics (Supplementary Material 1-A, mean = 61.95 years,
 257 SD = 11.47 years) suggests a slight skew. A histogram and box plot (Figure 2A and
 258 2B) were generated to corroborate this information (Q-Q plot, Supplementary

259 Material 2-A). Age has a very slight left-tailed skew, though it is not to an extreme
260 that would harm our final model, nor does it affect ANOVA testing due to sample
261 size ($n = 1643$). The statistical summary was faceted by sex (Supplementary
262 Materials 1-B and 2-B), which did not show notable differences in shape, median,
263 mean or quartiles, although a binomial test shows a statistical difference in
264 proportion (59.2% male, $p = 1.16e-13$, 95% CI = 0.567-0.615). This is sensible, as
265 males, are more often effected by CRC than females [2, 6] it would be expected for
266 this skew to be present in studies regarding CRC. An ANOVA found a statistical
267 difference in the mean ages of the groups complementary to the box plot (95% CI,
268 $p < 6.85e-11$, Supplementary Material 3-A), requiring a follow-up Tukey's HSD
269 test to determine which pairings were significantly different from each other. The
270 following pairings were found to be statistically different: HC–Other, PA+–HC,
271 CRC–HC, CRC+–HC, CRC+–CRC, and CRC-H–CRC+ (See Supplementary
272 Materials 3-A for individual p-values). All other mean differences in age showed
273 no statistical significance at a 95% confidence interval.

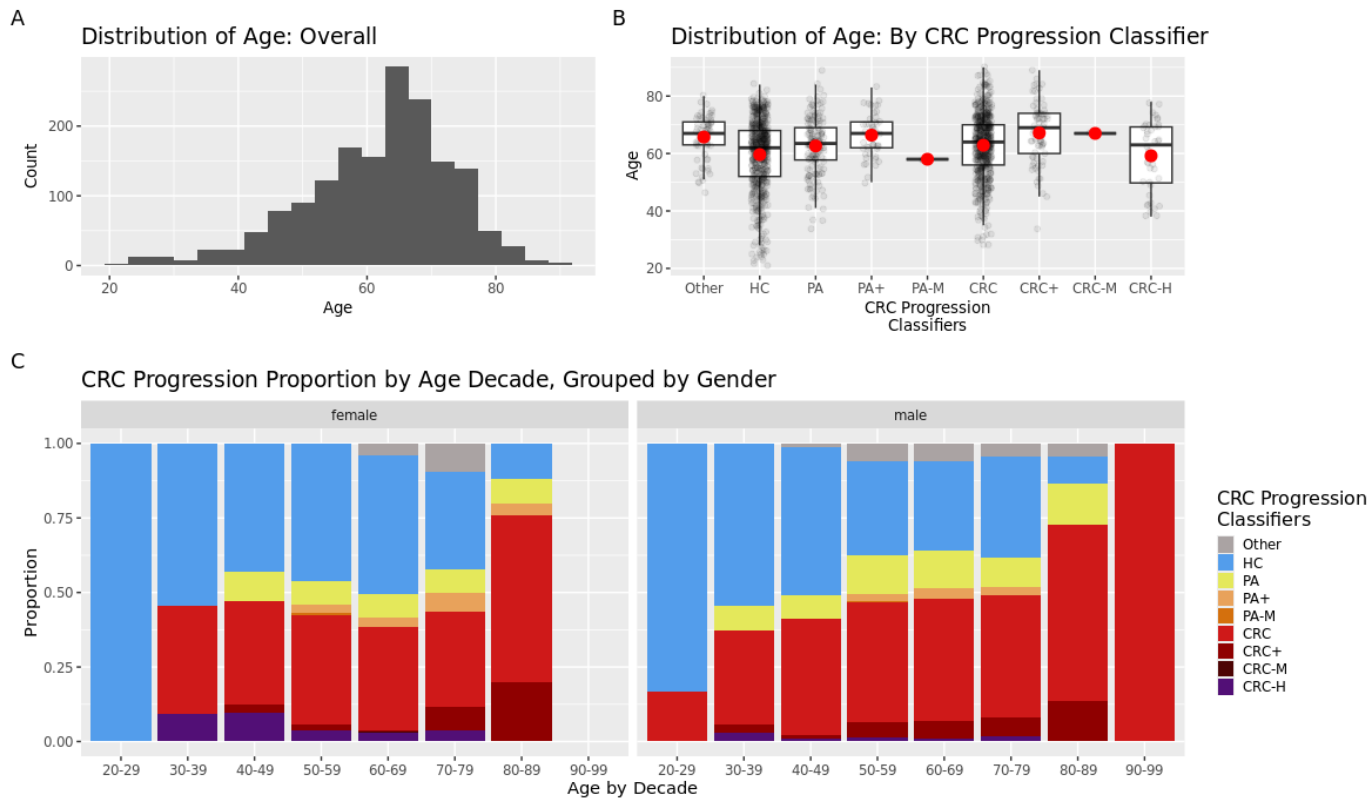


FIG 2 (A) Histogram demonstrating age distribution with a peak near 65 years. (B) Box plot of age across CRC progression classifiers, with jittered translucent points to demonstrate spread of the sample. The red dot represents the mean, the line represents the median. (C) Bar plot displaying the gender faceted proportion of CRC progression classifiers across each decade. Notable differences include and visual increase in HC from 20–69 for females, and increased PA and CRC/CRC+ across most decades in males.

The age, divided into decades, shows the three most represented groups are 60-69, 70-79, and 50-59 (34.4%, 21.8%, 19.9%, respectively; Supplementary Material 1-C). Figure 2C represents this in proportions for easier comparisons of the CRC

278 progression classifiers across the decades, faceted by gender, as it is well
279 established that males have a higher relative risk of CRC than females [2, 6]. This
280 pattern is reflected across most age groups in our data, particularly from 30-79.
281 Furthermore, visually there are less samples with recorded comorbidities, as well
282 as less PA compared to CRC and HC, overall. The three most affected countries by
283 CRC are well represented in this multi-cohort study (Supplementary Material 1-D
284 and 2-C), and within these three target countries, there is representation from 6 of
285 our classifiers: Other, HC, PA, CRC, CRC+, and CRC-H. The only unrepresented
286 classifier is PA+.

287 The normality of BMI was also checked using both a histogram for overall
288 spread and a box plot for distribution of BMI among the CRC progression
289 classifiers (Figure 3A and 3B, respectively). With a slight right-tailed skew and a
290 few outliers, particularly in the upper BMI bounds of the PA, PA+, and CRC, there
291 are visually striking differences among the means. An ANOVA (Supplementary
292 Material 3-B) showed statistical significance (95% confidence, $p < 2e-16$),
293 warranting a Tukey's HSD for further investigation. Of 36 pairings, 12 were found
294 to be statistically significant to a 95% confidence level, meaning BMI could have
295 an impact (Supplementary Materials 3-B). BMI and age were also plotted against
296 one another, shown in Figure 3C with the associated linear models for each CRC
297 progression classifier represented on the scatter plots (Supplementary Material
298 3-C). Results of the age-centered, linear regression modeling shows a very low
299 multiple R-squared value of 0.0786, meaning that BMI and age only explained
300 about 8% of the variance, though there are a few relationships that show statistical
301 significance to a 95% confidence level.

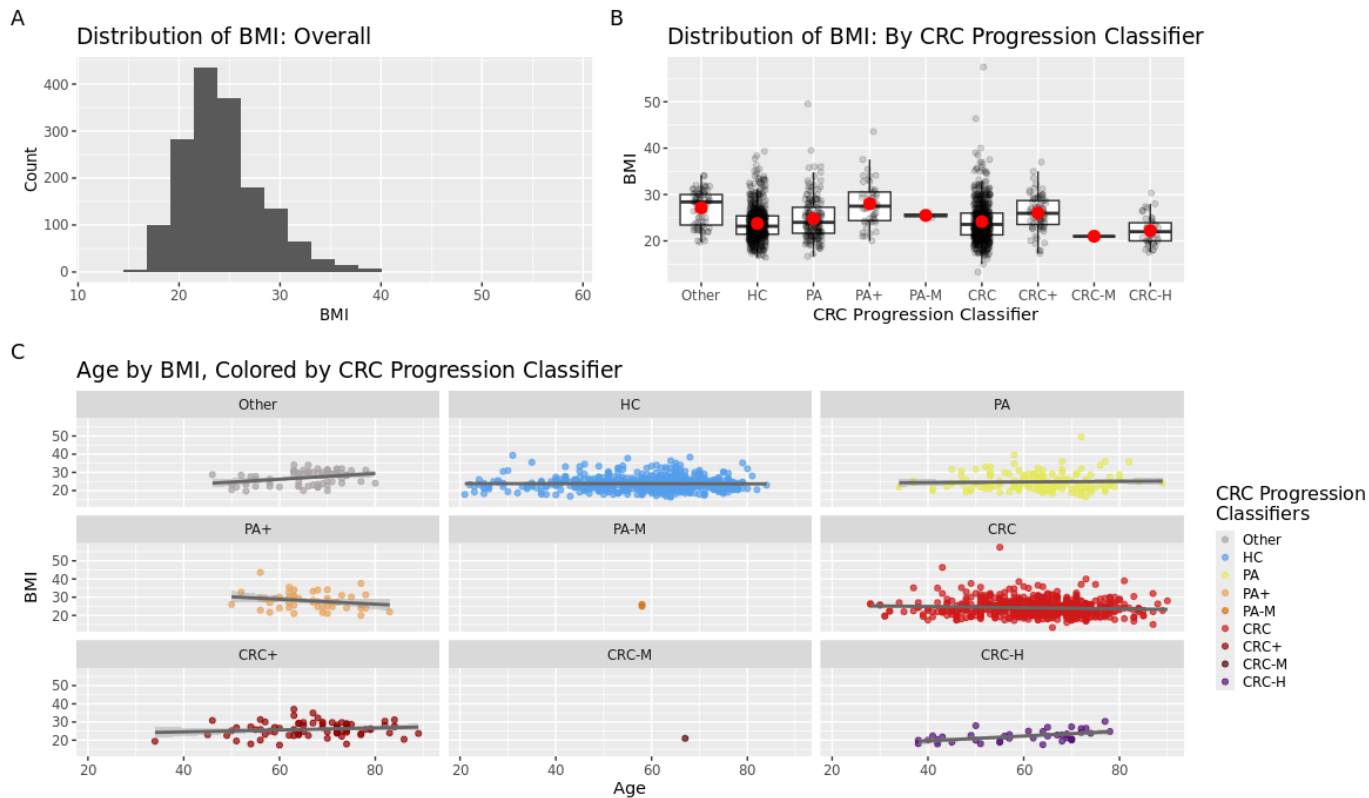


FIG 3 (A) Histogram showing distribution of body mass index (BMI) across samples with a peak around 23. (B) Box plot of BMI by CRC progression classifier. The red dot marks the mean, while the black line marks the median. All classifiers generally have a normal distribution. (C) Scatter plots of age by BMI, with a linear model regression line of y by x , faceted by CRC progression classifier. Notice that there is minimally notable clustering.

303 **Taxonomic and Pathway Analysis**

304 To identify any trends in the data, we looked at beta-diversity, which is a test
305 for dissimilarity between the study conditions. Although the p-value was highly
306 significant ($p = 0.001$), the R-squared value ($R^2 = 0.0078$) indicates much of the
307 dissimilarity is not explained by the study conditions. These results lead us to
308 investigate mitigation strategies for potential batch effects from the studies
309 themselves that could muddle results.

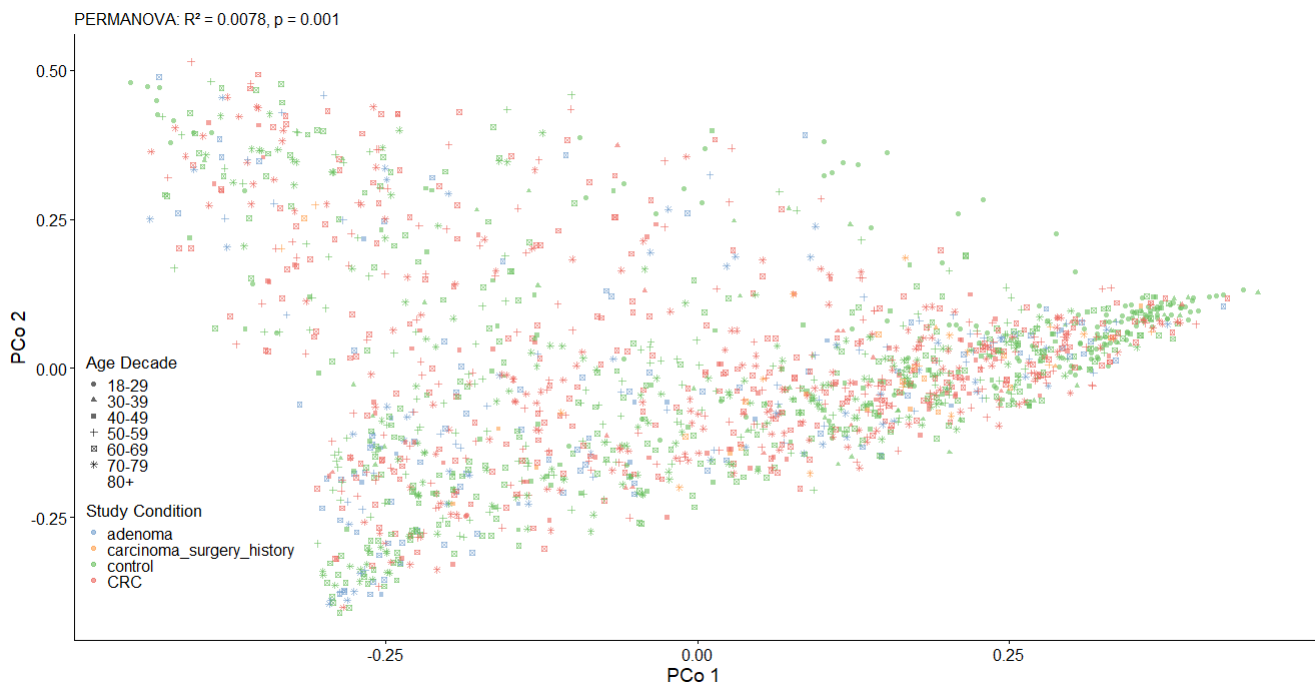
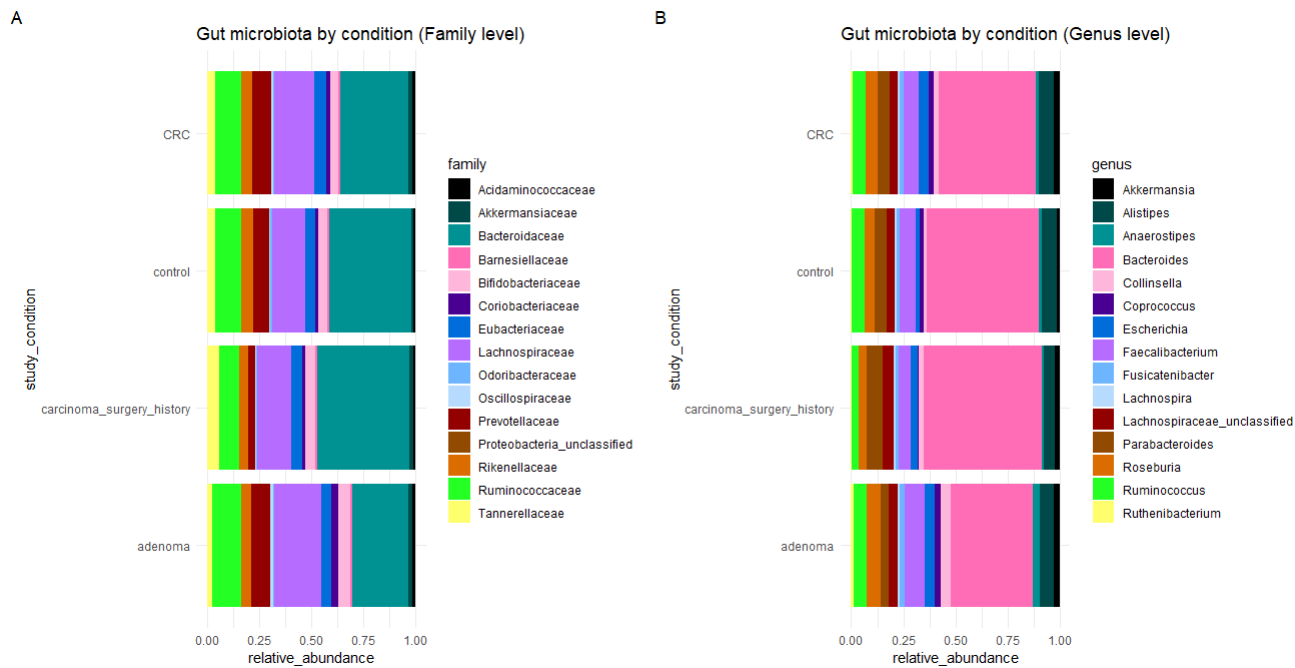


FIG 4 Principle Coordinate Analysis illustrating Bray-Curtis dissimilarity of metagenomic data from cMD3 study. Comparisons were made between study conditions and age groups.

311 To understand how the taxa are distributed across the different study
312 conditions, two relative abundance plots were generated at the family and genus
313 level.



314

FIG 5 Stacked bar plots illustrating relative abundance of cMD3 metagenomic data based on study conditions of filtered studies (A) Family taxonomic level (B) Genus taxonomic level. study_condition is the class by which the original study classified the participants as during investigation. They would closely correspond to HC, CRC, PA, and CRC-H.

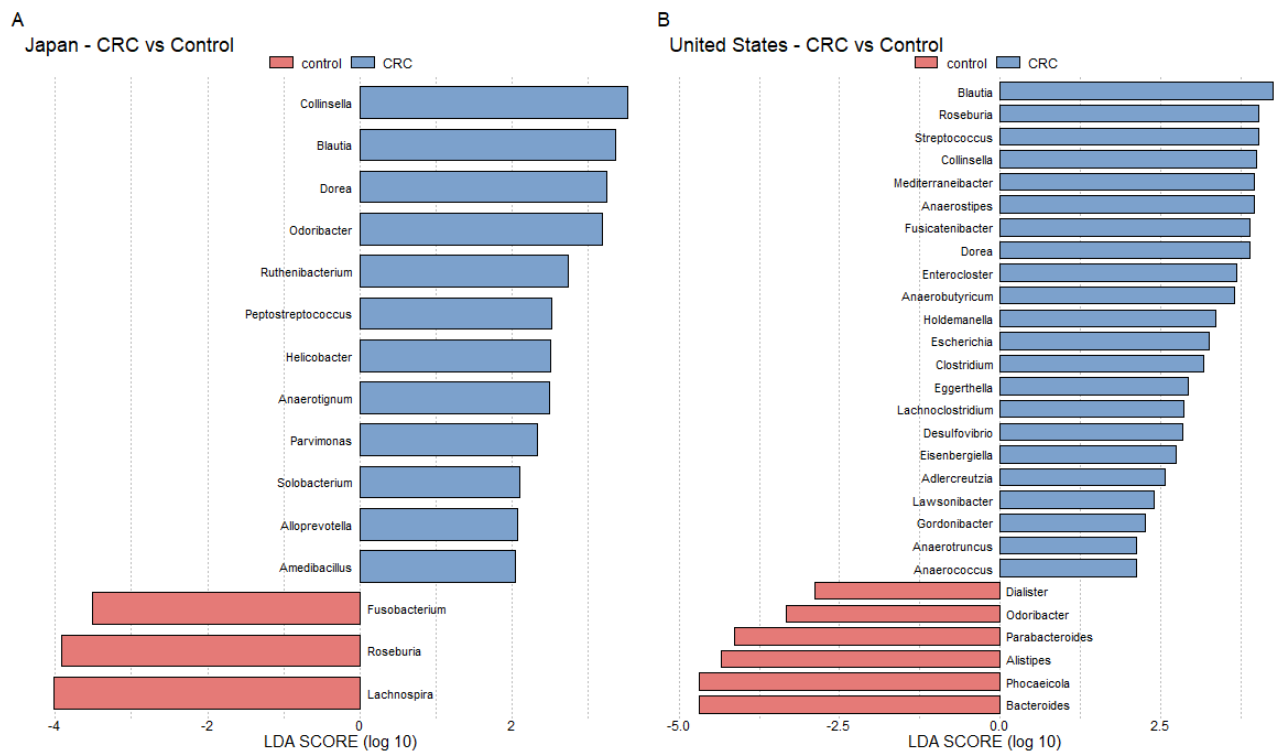


FIG 6 Differential abundance of cMD3 Study metagenomic data using LDA Effect Size (LEfSe) comparing healthy controls and subjects with colorectal cancer (CRC) in (A) Japan and the (B) United States.

study_condition is the class by which the original study classified the participants as during investigation. They would closely correspond to HC, CRC, PA, and CRC-H.

The relative abundance stacked bar plots (Figure 5) also indicate there was very little difference at the family and genus taxonomic levels. Considering these data sets cover several countries and age categories, it is possible that further separation into more groups may provide better comparisons. As Japan and the U.S. contain the greatest number of participants, these countries were separated out. For the same reason, only the control and CRC groups were compared, as

these conditions have the highest sample sizes. To identify key differences in how taxa may be enriched between groups, differential abundance was looked at using LEfSe analysis (Figure 6). Using a reference (control) and a treatment (CRC), LEfSe performs Kruskal-Wallis test to identify taxa that were statistically different from zero between the groups. Taxa found to be statistically significant were compared pairwise with a Wilcoxon Rank-Sum Test. Finally, the effect size was calculated using LDA. Three genera (*Blautia*, *Collinsella*, and *Dorea*) were found to have greater abundance in CRC samples compared to controls, which were compared to the top 20 features of final models.

Random Forest Models:

While there was desire for keeping a multi-kingdom model for increased accuracy, due to the prevalence filtering applied during preprocessing, the resulting filtered matrix contained only bacterial species. Furthermore, despite initial trials separated HC, PA (binned PA and PA+), CRC (binned CRC and CRC+), and Other, particularly when running 80/20 split models (Supplementary Materials 5), binning the RF classifiers into POSITIVE_POLYP and NEGATIVE_POLYP according to Table 2 classes gave more balanced class weighting with superior accuracy, sensitivity, and F1 scores, outlined below.

Model 1: 80/20 Test Train Split

The confusion matrix of the 80/20 test train split model had an overall accuracy of 0.6489, with the strongest performance classifying the positive polyp class. Table 6 summarizes the results of this model, with Figure 7A displaying the confusion matrix. The model yields a p-value of 0.0026, indicating statistically

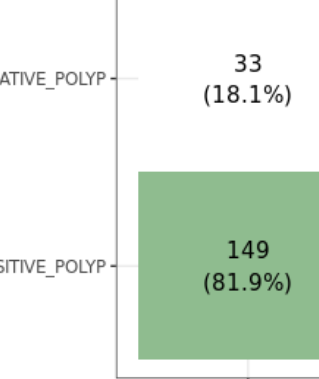
345 significant classification performance at the 95% confidence level. Important to
346 note is the low false negative rate and high false positive rate, reflected in the
347 sensitivity and specificity. The top 20 most important features used in the model
348 generation is available in Supplementary Materials 6. Of these top 20 features,
349 20% are taxonomic abundance, 60% are pathway abundance, 20% are pathway
350 coverage, and no metadata metrics. Three of the four taxonomic abundance are in
351 the top three most important features: *Gemella mobillorum*, *Peptostreptococcus*
352 *stomatis*, and *Solobacterium moorei*. Interestingly, the three previously discovered
353 genera from the LEfSe analysis, *Blautia*, *Collinsella*, and *Dorea*, were not among the
354 top 20 features, despite being the statistically significant genera from the two
355 largest countries.

Metric	Value
POSITIVE_POLYP (n_{train})	730
NEGATIVE_POLYP (n_{train})	551
OOB Prediction Error	0.2140
Accuracy	0.6489
Balanced Accuracy	0.6210
P-Value	0.0026
Kappa	0.2525
Sensitivity	0.8187
Specificity	0.4234
Positive Predictive Value	0.6535
Negative Predictive Value	0.6374
F1 Score	0.7268
POSITIVE_POLYP AUC	0.7170
NEGATIVE_POLYP AUC	0.7170
Overfit Check	0.3511

TABLE 6 Percentile normalized model 80/20 split results.
 $Overfit_Check = Accuracy_{Train} - Accuracy_{Test}$.

A

Confusion Matrix: 80/20 Split

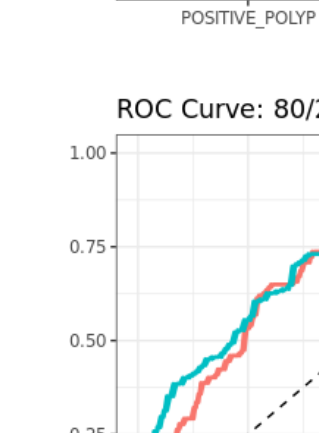


The confusion matrix shows the relationship between predicted and actual classes. The y-axis represents the predicted class (NEGATIVE_POLYP, POSITIVE_POLYP) and the x-axis represents the actual class (POSITIVE_POLYP, NEGATIVE_POLYP). The cells contain the count and percentage of instances.

Predicted \ Actual	POSITIVE_POLYP	NEGATIVE_POLYP
NEGATIVE_POLYP	33 (18.1%)	58 (42.3%)
POSITIVE_POLYP	149 (81.9%)	79 (57.7%)

B

ROC Curve: 80/20 Split



The ROC curve plots the True Positive Rate (Y-axis) against the False Positive Rate (X-axis). The red line represents the NEGATIVE_POLYP class and the blue line represents the POSITIVE_POLYP class. A dashed diagonal line represents random performance.

True Positive Rate

False Positive Rate

Class

- NEGATIVE_POLYP
- POSITIVE_POLYP

25

358 *Model 2: Leave-One-Dataset-Out Cross-Validation*

359 The LODO cross-study had a wide range of performance, with the highest
360 balanced accuracy in ThomasAM_2019_c (0.763) and ThomasAM_2018a (0.740).
361 Lower performance was observed in GuptaA_2019 (0.520) and FengQ_2015
362 (0.613). Study specific kappa analysis showed moderate agreement across datasets
363 with the highest in ThomasAM_2019_c (0.525), with ThomasAM_2018b following
364 closely behind (0.411). GuptaA_2019 showed extremely low agreement at 0.033,
365 seen in Table 7. The sensitivity was often high and specificity low (0.75–1 and
366 0.03–0.52, respectively), with specificity clustering around 0.38-0.59. A faceted
367 AUC-ROC is available in Supplementary Materials 7, where the variety in
368 performance across the held-out study models is clear.

369 *Results Summary*

370 The models results in Tables 6, 7, and 8 demonstrate strong performance for
371 the identification of a positive case via the confusion matrix and sensitivity, but
372 limited ability to consistently detect negative cases via the specificity. The 80/20
373 random split gives overly optimistic performance, while LODO provides a more
374 realistic estimate of external generalizability across studies and an estimate of real
375 world performance on unseen datasets.

376 Notably, the observed reductions in specificity of the LODO model likely
377 reflects increased heterogeneity in the globally defined healthy class, leading to
378 misclassifications of healthy samples as diseased. This differs from many prior
379 studies that rely on internal comparisons [30]. The use of a global healthy control
380 introduces greater variability and more closely reflects real world diagnostic
381 conditions across diverse populations.

Hold-Out Study	n_{POS}	n_{NEG}	Accuracy	Bal Accuracy	P-Value	Kappa
FengQ_2015	819	627	0.6494	0.6222	0.1419	0.2501
GuptaA_2019	882	658	0.5167	0.5167	0.4487	0.0333
HanniganGD_2017	859	660	0.5556	0.5425	0.9749	0.0795
ThomasAM_2018a	859	664	0.7403	0.6631	0.1957	0.3502
ThomasAM_2018b	880	661	0.7119	0.7025	5.833e-03	0.4110
ThomasAM_2019_c	872	648	0.7625	0.7625	1.366e-06	0.5250
VogtmannE_2016	860	636	0.5192	0.5192	0.3844	0.0385
WirbelJ_2018	852	623	0.6320	0.6391	7.533e-03	0.2740
YachidaS_2019	582	442	0.6597	0.6306	1.252e-05	0.2731
YuJ_2015	838	634	0.7109	0.6674	1.318e-03	0.3610
ZellerG_2014	817	627	0.6090	0.5733	0.5350	0.1516

TABLE 7 Percentile normalized model LODO results pt.1. n_{POS} = number of POSITIVE_POLYP samples in training split. n_{NEG} = number of NEGATIVE_POLYP samples in training split. Bal Accuracy = Balanced Accuracy.

Hold-Out Study	Sensitivity	Specificity	Pos Pred Val	Neg Pred Val	F1 Score	AUC_{POS}	AUC_{NEG}
FengQ_2015	0.7527	0.4918	0.6931	0.5660	0.7216	0.6734	0.6734
GuptaA_2019	1.0000	0.0333	0.5085	1.0000	0.6742	0.8111	0.8111
HanniganGD_2017	0.5849	0.5000	0.6889	0.3889	0.6327	0.5155	0.5155
ThomasAM_2018a	0.8679	0.4583	0.7797	0.6111	0.8214	0.7036	0.7036
ThomasAM_2018b 383	0.8125	0.5926	0.7027	0.7273	0.7536	0.7847	0.7847
ThomasAM_2019_c	1.0000	0.5250	0.6780	1.0000	0.8081	0.9550	0.9550
VogtmannE_2016	0.7885	0.2500	0.5125	0.5417	0.6212	0.6239	0.6239
WirbelJ_2018	0.8167	0.4615	0.5833	0.7317	0.6806	0.7979	0.7979
YachidaS_2019	0.8303	0.4309	0.6618	0.6543	0.7366	0.6970	0.6970
YuJ_2015	0.9459	0.3889	0.6796	0.8400	0.7910	0.8301	0.8301
ZellerG_2014	0.7368	0.4098	0.6604	0.5000	0.6965	0.6928	0.6928

TABLE 8 Percentile normalized model LODO results pt.2. Pos Pred Val = Positive Predictive Value. Neg Pred Val = Negative Predictive Value. AUC_{POS} = POSITIVE_POLYP AUC. AUC_{NEG} = NEGATIVE_POLYP AUC.

384 **DISCUSSION**

385 **Key Findings**

386 As CRC diagnoses continue to increase worldwide, there is an imperative for
387 non-invasive tools to assist with early detection. This investigation sought to build
388 a foundational proof-of-concept machine learning tool via a large publicly
389 available dataset for classifying CRC progression based on microbial
390 metagenomics from fecal samples. With the 80/20 split AUC at 0.7170, and the
391 statistically significant LODO models' AUC results falling between 0.6970 and
392 0.9550, our models are well within the bounds of those found within the current
393 literature. Studies investigating CRC detection using microbiomes tend to reflect
394 wide variance in predictive accuracy. The AUC range of T. Li et al. [19] (0.585–0.88)
395 and Manghi et al. [16] (0.51–0.95) encompass our results. Since our data comes
396 directly from the database generated by Manghi et al. [16], it is expected that our
397 models most closely resemble the AUC values from their paper. Interestingly,
398 when separated into the HC, PA, CRC, and Other classes (Supplementary
399 Materials 5), the per-class AUC results are between 0.6798 and 0.8836, also falling
400 within these boundaries. This separated version was not pursued due to the
401 desire to identify the early PA cases; the sensitivity of PA is 0.0000 in that model
402 (Supplementary Materials 5-A), meaning these early cases were not correctly
403 identified. If they were misclassified as CRC, further screening would still be
404 initiated; however, Supplementary Materials 5-C shows that more than half are
405 misclassified as HC. This difficult classification is likely due to the intermediate
406 nature of these polyps with a wide range of stages, potentially reflected in the
407 metagenomes and causing fingerprinting to be more problematic. Since early

408 detection remains one of the best methods for positive patient outcomes [2], it was
409 decided to continue with the binning method outlined in Table 2.

410 The relatively high sensitivity of both the binary Models 1 and 2 shows they
411 are adept at classifying the majority of positive polyp cases, corroborated by the
412 confusion matrix plotted in Figure 7. This high sensitivity is ideal in cases for
413 which missing the disease, or a false negative, is risky [31]. However, the low
414 specificity means that many of the negative cases falsely classified as a positive
415 case. To determine model readiness, we decided it would be best determined
416 through comparisons to an in-use clinical test, for which FOBT, another
417 non-invasive polyp screening method, was used. FOBT has a wide range of
418 diagnostic accuracy, sensitivity, and specificity. Lu et al. [8] found that the
419 accuracy (0.10-0.20), sensitivity (0.60-0.72) and specificity (0.88-0.95) for detecting
420 advanced adenomas and CRC ranges widely depending on the subtype of FOBT,
421 testing the accuracies across different growth locations within the colon. Their
422 summary ROCs ranged from 0.531-0.942 dependent on tumor stage and location
423 [8]. Ramdzan et al. [32] found a wider range of results in their meta-analysis of
424 FOBT accuracy, with sensitivity at 0.31-0.71 and specificity at 0.87-0.95. Although
425 the accuracies and sensitivities of our models (Tables 6-8, Supplementary
426 Materials 5) are well within or outperform the range of values for accuracy and
427 sensitivity of FOBT, the specificity does not. Compounding on this is a known
428 overfitting issue within the current set-up, suggesting the need for further data
429 collection, tuning, and refinement before further processing into a patient-ready
430 diagnostic tool. However, these results are incredibly promising for the realization
431 of a metagenomic microbial CRC diagnostic tool.

432 Within this potential diagnostic tool, the model features (Supplementary
433 Materials 6) paired with a healthcare provider's expertise could provide insights

434 to the patient's health or aid in differential diagnosis and treatment. Recent
435 studies found the genera *Gemella* and *Peptostreptococcus* were both more abundant
436 in mucosal samples from patients with CRC [33, 34], and fecal enrichment of
437 *Solobacterium moorei* in CRC patients with its contribution to CRC progression
438 promotion in model organisms [35]. This finding is promising, as the first three
439 features in our model were of these species. Additionally, pathway abundance
440 and coverage would be most beneficial for future treatment and research avenues.
441 A growing body of evidence suggests targeting specific anaerobic microbial
442 pathways may be useful for treatment against cancerous cells in the colon,
443 particularly the pyruvate fermentation pathway found in our feature table [36].
444 This finding, along with other valuable patient history, may help form a functional
445 profile to more accurately detect early signs of polyps and build potential
446 treatment targets.

447 **Limitations**

448 Several limitations remain related to data availability, metadata completeness,
449 and necessary inclusion criteria. The aggregated nature of the dataset brings major
450 consideration into the batch effects for the analysis, specifically the difference in
451 laboratory techniques, sequencing machines, extraction kits, and protocols used
452 throughout all of the studies. While documented and partially corrected for, they
453 were not fully corrected, thus requiring mention as they create biological noise in
454 the modeling. These differences between studies also creates the major limitation
455 of incomplete availability of clinically and biologically relevant metadata; this is
456 not unique to our analysis, but instead reflects broader challenges in large-scale
457 meta-analyses, as gaps in metadata is a shortcoming impacting the field of ML in
458 metagenomic studies as a whole, affecting both modeling and interpretations [37].

459 In our case, despite the cMD3 containing over 30 features of accumulated
460 metadata, the majority are incomplete due to methodology differences across
461 studies (Figure 1), rendering them irrelevant to the model; variables that are
462 known to strongly influence the microbiome, such as antibiotic use, smoking
463 status, alcohol consumption, and disease stage [38], are inconsistently reported or
464 missing. Other important indicators of CRC, such as FOBT results, are also
465 consistently missing, which could have been important as a predictor in the
466 model. Only a subset of variables are consistently available across the dataset,
467 such as sequencing platform, participant country, gender, age, BMI, and disease
468 class. Furthermore, there are entire countries and continents missing from the
469 dataset, such as African and South American countries, suggesting a need to
470 include these already underrepresented populations. Many use cases of the
471 proposed proof-of-concept tool will need to be deferred until more complete
472 datasets become available, or studies are tailored to collect more relevant data for
473 development.

474 Related to the missing metadata are disease subclassification; while this study
475 focuses on CRC metagenomic analysis/fingerprinting, the available data has
476 either empty or missing columns for features such as tumor stage, location, and
477 metastasis. Original intentions were to separate RF classifiers into the full range of
478 possible diagnoses, such as PA, PA+, PA-M, CRC, CRC+, and CRC-M, which is
479 currently not feasible until more complete data are available. This will result in a
480 computational framework that can only compare broad disease categories rather
481 than detailed cancer stages. With the binning of samples, the PA category
482 (including PA+, excluding PA-M) was not correctly identified at all. Data filtration
483 for data quality limited the data set size, creating a working database dominated
484 by HC and CRC samples where polyp-only and adenoma cases are few. Due to

485 this, disease progression from polyp to cancer modeling was not reliable, leading
486 to a focus on a binary classification system. This binary classification system also
487 appeared to inadvertently increase the overfitting in the system compared to the
488 multi-class attempt, despite hyperparameter tuning and other efforts to reduce
489 overfitting. With higher sample size and better class balance, overfitting may be
490 better addressed.

491 Given the relatively limited sample size after filtering, we explored both a
492 conventional 80/20 split and a LODO cross-validation approach. While the 80/20
493 split, performed in a study-aware manner, was useful for faster model tuning and
494 iteration, it still allows for subtle patterns in study-specific characteristics between
495 training and testing data, causing the model to learn batch effects or technical
496 artifacts rather than a true biological signal. In contrast, the LODO framework
497 provides a more stringent evaluation by holding out entire studies, highlighting
498 how performance can degrade when the model is exposed to completely unseen
499 cohorts. Our results underscore the risk of overfitting in aggregated microbiome
500 datasets and reinforces the importance of LODO as a more reliable estimate of
501 real-world generalization, even if it is computationally more expensive. However,
502 due to the limited number of repeated countries, LODO inadvertently removes
503 entire countries from the training data, which becomes problematic as it is well
504 known that country, and consequently diet, are significant influences on the gut
505 microbiome [39]. The LODO has shown substantial variation, suggesting limited
506 generalizability at this stage, further supporting the need for additional datasets.

507 **Future Research and Recommendations**

508 With expansions of the cMD3 [16, 40], and scripts available for further
509 integration and harmonization into the database framework, increasing the
510 number of datasets could improve accuracy and reduce overfitting. A search for
511 studies that contain PA patient samples to balance the dataset for strengthening
512 the multi-class model is particularly important, especially as Supplementary
513 Materials 5 shows the multi-class system has a reduced overfitting compared to
514 the binary model. This would better capture early stages of CRC, improving
515 patient outcomes. While this model is tailored to the cMD3 data and selected
516 studies, generalization of the written code allows researchers to apply these
517 models to studies of various microbiome data such as oral and vaginal, among
518 others. Future tuning and applications to these different sample locations are at
519 the forefront of ongoing research planned within the group.

520 In conjunction with inclusion of future studies into the model matrices, new
521 and ever changing modeling techniques can be applied, namely ensemble
522 learning. As seen in the Naderalvojud et. al. study [41], ensemble learning can be
523 implemented to improve accuracy and reliability in ML models. With this, a
524 boosting approach is used, where further models are developed to focus on fitting
525 misclassified samples. Our models face this misclassifications problem,
526 particularly with the negative polyp class and with the precursor model where PA
527 was entirely misidentified; ensemble learning may resolve some of these issues.
528 Confounding factors may still remain at play, and more robust ways to mitigate
529 them should also be explored. One option is from Chyzhyk et al. [42], whom
530 apply a cross-validation isolation procedure that allows accurate predictions and
531 safe interpretations regardless of confounding effects. Another option is from Liu

et al. [43], whom developed miMatch to eliminate individual metabolic background differences to reduce false positives and increase generalizability, which again is something these models have issues with. It is also important to note that studies with samples from countries and continents currently missing in the cMD3 data are of high importance to future works as well.

For clinical testing, it is known that CRC incidence is on the rise in younger individuals [22]. If metagenomic testing becomes a viable diagnostic tool with high sensitivity capable at detecting positive cases, it could be prudent to test adults using these non-invasive methods at earlier ages, prior to the recommended colonoscopy screening age, in conjunction with other non-invasive screening methods for boosted predictive value. This way, early detection becomes more feasible with better outcomes for patients, reducing the burden of cancer on the population.

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DATA AVAILABILITY STATEMENT

All data and code used in this study is available via the study's github repository found in Supplementary Materials 8-A.

552 **ETHICS APPROVAL**

553 The authors declare that no ethical approval is needed for this research.

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559 **CONFLICTS OF INTEREST**

560 The authors declare no conflict of interest.

561 **SUPPLEMENTAL MATERIAL**

562 Please see attached supplemental materials.

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